

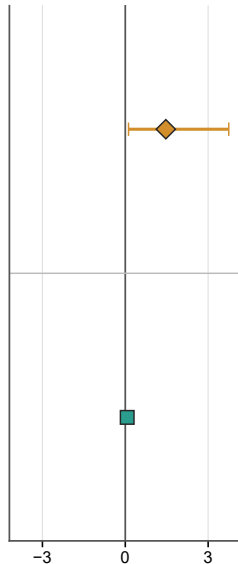
Improvement No change Deterioration

(a) Lateral-error RMS reduction (m)

P1	6.86
P2	1.25
P3	0.664
P4	1.39e-04
P5	0
P6	0.0385

P1	0.0449
P2	0.329
P3	0.0414
P4	-6.03e-04
P5	-2.93e-04
P6	1.71e-04

Mean effect (95% CI)

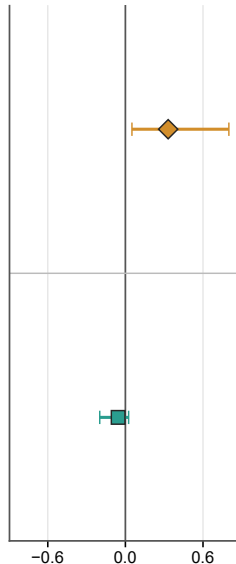


(b) Heading-error RMS reduction (rad)

P1	1.48
P2	0.202
P3	0.162
P4	3.91e-04
P5	0
P6	0.141

P1	-0.403
P2	0.0481
P3	0.0123
P4	-2.71e-04
P5	-5.55e-04
P6	6.40e-03

Mean effect (95% CI)

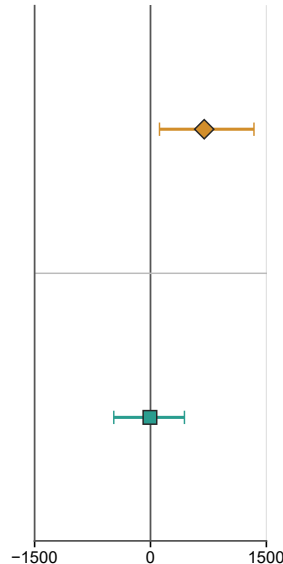


(c) Input-increment index reduction ($I_{\Delta u}$)

P1	1931
P2	1543
P3	699
P4	-0.291
P5	0
P6	-0.173

P1	-964
P2	879
P3	41.9
P4	-0.0252
P5	1.12e-03
P6	4.31e-03

Mean effect (95% CI)



Truth-driven reference

Causal-IMU

Cells show zero-grade comparator minus target (absolute route effects); positive values favor the target. Color saturation is scaled independently within each metric.